

Report

Meteorological Drought Analysis over Peninsular India

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Study Area: Peninsular India

Data: Precipitation (monthly (Jan 2000 – September 2024))

Probability distribution used: Gamma Distribution.

Steps Followed for (SPI Analysis):

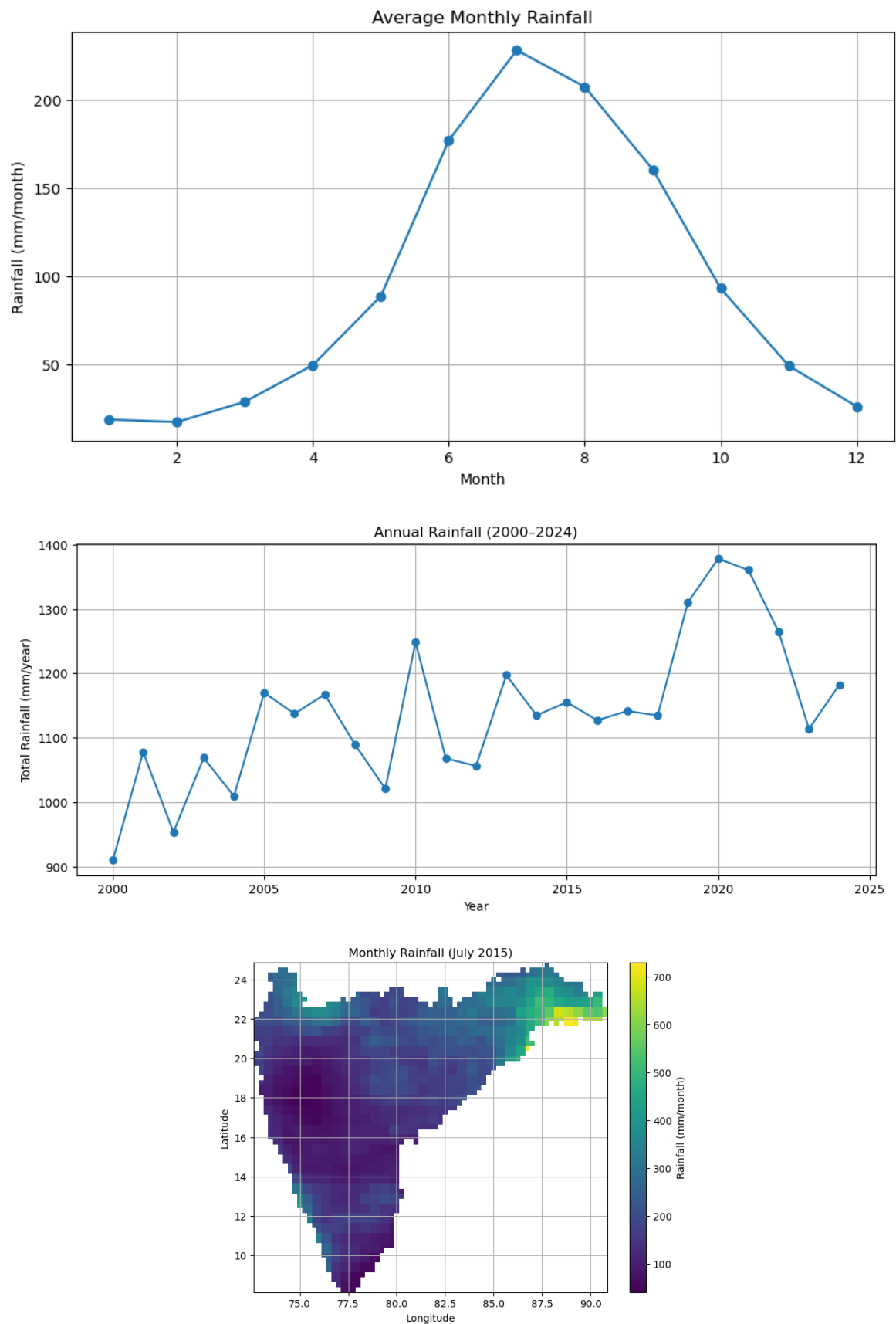
1. Used monthly precipitation data (2000–2024) in NetCDF format for the study.
2. Selected Peninsular India as the region of interest (ROI).
3. Extracted rainfall data for all grid cells within the study region.
4. The dataset provided precipitation in **mm/day** (monthly mean). Therefore, it was converted to monthly cumulative rainfall (**mm/month**) using the number of days in each month.
5. Prepared a continuous monthly rainfall dataset, excluding missing months (Oct–Dec 2024).
6. Computed 3-month and 6-month rolling rainfall totals to obtain SPI-3 and SPI-6.
7. For each grid cell (pixel), fitted the rainfall series to a Gamma probability distribution.
8. Converted the fitted distribution to cumulative probability (CDF).
9. Transformed the CDF values to standard normal values to obtain SPI (mean = 0, standard deviation = 1).
10. Identified drought conditions at each pixel using the threshold $SPI < 0$.
11. Extracted drought events and computed drought characteristics (frequency, duration, severity, and intensity) for each pixel.
12. Generated spatial maps of all drought metrics for both SPI-3 and SPI-6.
13. Computed spatial averages of the metrics to obtain the regional values.
14. Compared drought behaviour between SPI-3 and SPI-6.

Note: SPI is based on accumulated precipitation over a time scale, not on average rainfall. If monthly mean rainfall (mm/day) is used, it represents average daily intensity, not total water availability. It does not reflect the actual moisture input to the system.

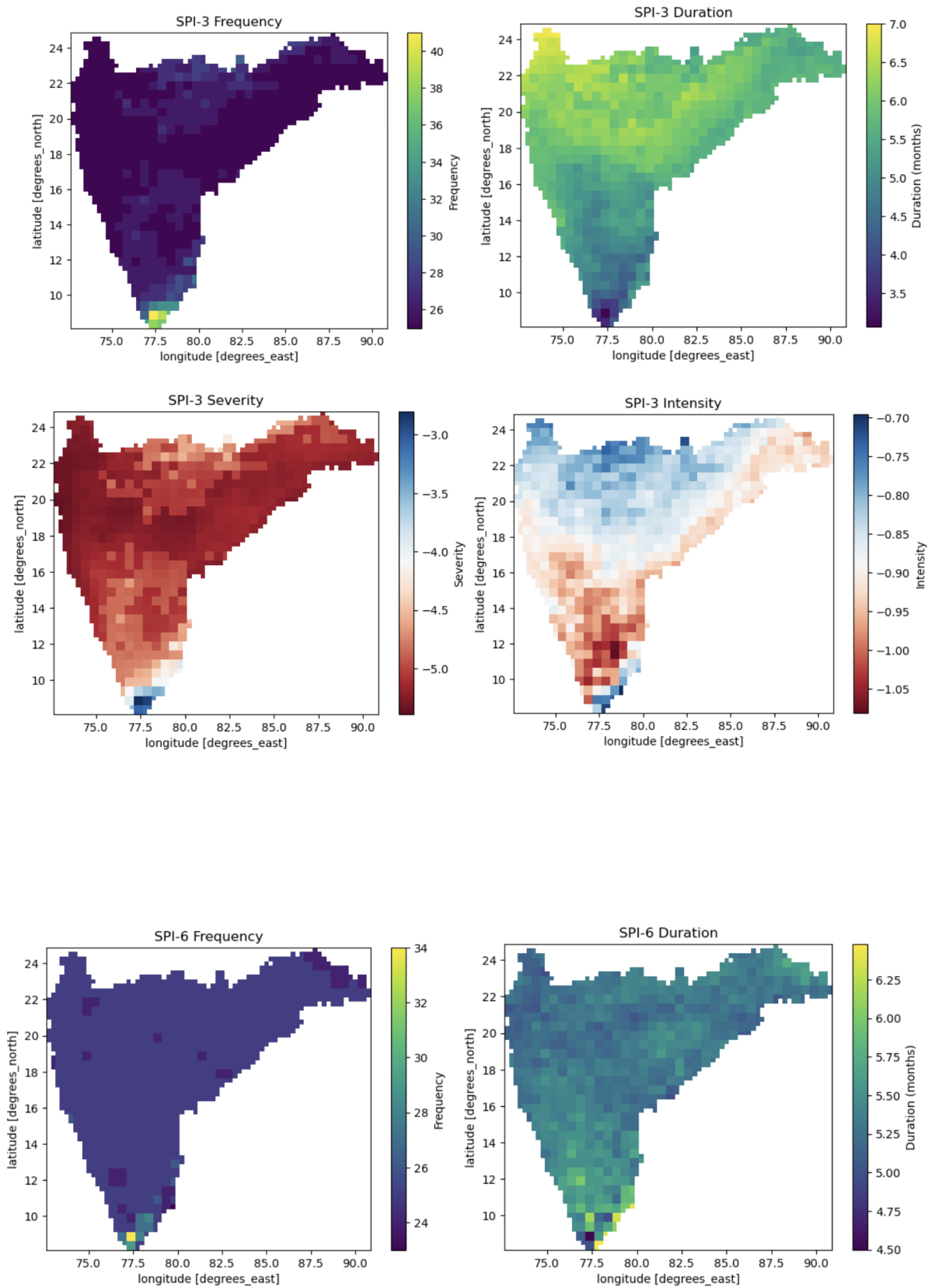
But SPI requires total rainfall received over a period (mm), because drought depends on the deficit of accumulated water, not the daily rate.

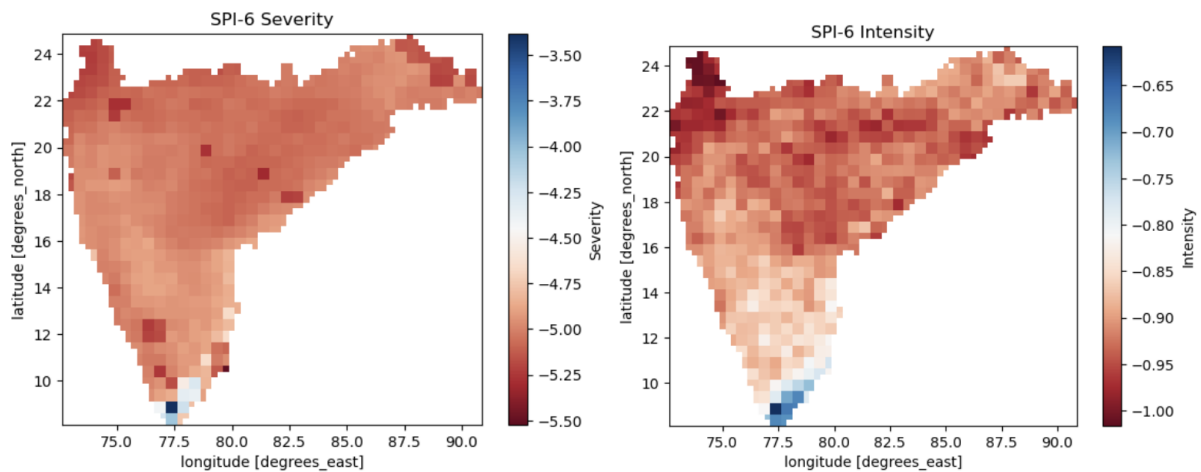
Reference: McKee et al. (1993), “Assessing the suitability of McKee et al. (1993) drought severity classification across India”, defined SPI using accumulated precipitation over specific time scales (e.g., 3-month, 6-month), which are then fitted to probability distributions and transformed to a normal distribution. DOI: [10.1007/s11069-024-06762-3](https://doi.org/10.1007/s11069-024-06762-3)

Pre-analysis results:



SPI Analysis Results [ASSIGNMENT]:





	Frequency	Avg Duration (months)	Avg Severity	Avg Intensity
SPI-3	25.680	5.692	-5.038	-0.883
SPI-6	25.002	5.348	-5.007	-0.911

Comment on drought properties such as duration, frequency, intensity and severity of meteorological droughts.

Basic Definitions of Drought Properties:

- 1. Drought Frequency:** Drought frequency represents the number of drought events occurring over the study period. It indicates how often the SPI falls below the threshold ($SPI < 0$) and forms distinct drought events.
- 2. Drought Duration (D_e):** Drought duration is expressed in months and represents the period during which the SPI value remains continuously below the threshold level ($SPI < 0$). In other words, it is the time between the start and end of a drought event.
- 3. Drought Severity (S_e):** Drought severity indicates the cumulative deficit during a drought event. It is calculated as the sum of SPI values over the entire duration of the drought. More negative values represent greater rainfall deficiency.
- 4. Drought Intensity (I_e):** Drought intensity is the average magnitude of drought during an event. It is calculated as the drought severity divided by the duration. It represents how strong the drought is on a month-to-month basis.

Comment on observed drought properties:

Droughts are frequent in the region, as the results show. Both the SPI-3 and SPI-6 events occur at an average frequency of about 25. Thus, dry periods occur at regular intervals each year. According to the spatial maps, this frequency is roughly the same throughout the area, though the southern tip is slightly higher than the rest.

In terms of duration, droughts last for roughly 5 to 6 months. According to the SPI-3 map, the longer duration is apparent in the northern and central areas, while the shorter event

is visible in the southern area. When the northern areas start facing drought, it is likely to persist for a long duration. In SPI-6, the duration pattern is almost the same throughout the area, but it appears that durations are slightly higher in the southern region than in the north.

Since the severity values represent rainfall deficit, it is expected that they would be negative everywhere. The northern area has higher negative values than the southern tip, indicating a greater total accumulated deficit there. The southern part, despite experiencing droughts, has lower negative values, which results in a relatively less overall deficit.

The intensity exhibits a somewhat altered pattern. The southern portion in SPI-3 has a higher intensity of negative value thus showing more intense drought condition month-to-month. Unlike the northern region, which has less negative intensity, the drought in the southern region is stronger on a monthly basis, even though its duration is shorter. The maps reflect a clear distinction between duration and intensity.

In general, the region suffers frequent droughts. The northern regions suffer longer and more severe droughts whereas the southern regions suffer shorter but more intense drought conditions.

Comment on differences between the droughts identified using SPI-3 and SPI-6.

The major visible difference between SPI-3 and SPI-6 is the used time scale. This clearly reflects in the results.

Rainfall variations on short time span is captured by SPI-3. Due to this, it exhibited a longer average duration as well as slightly more negative severity. It also exhibits more differences as one moves across space, particularly in intensity. SPI-3 shows the southern part more distinctly has moderately intense monthly droughts.

In contrast, SPI-6 smooths out short-term fluctuations. The frequency is nearly the same, but the duration is somewhat lower and more uniform across the region. The negative severity values are somewhat less severe with respect to SPI-3, which indicates the spreading of the deficit over a longer time span. On average, however, the intensity in SPI-6 is somewhat more negative, indicating more consistently strong drought conditions over time.

The spatial pattern is yet another difference. SPI-3 indicates sharper differentiation between regions, especially north and south. SPI-6 delivers a more uniform pattern by combining short dry periods to create longer ones.

Put simply, the short-term SPI-3 is more volatile and varying when compared to the long-term SPI-6 which is often stable.